

Overview

Mode-transitioning robotic leg design to shift between high-speed and high-torque configurations for search and rescue applications.

Rescue Robot

- Legged robot to navigate rough terrain
- High Speed for rapid traversal
- High torque for casualty extraction

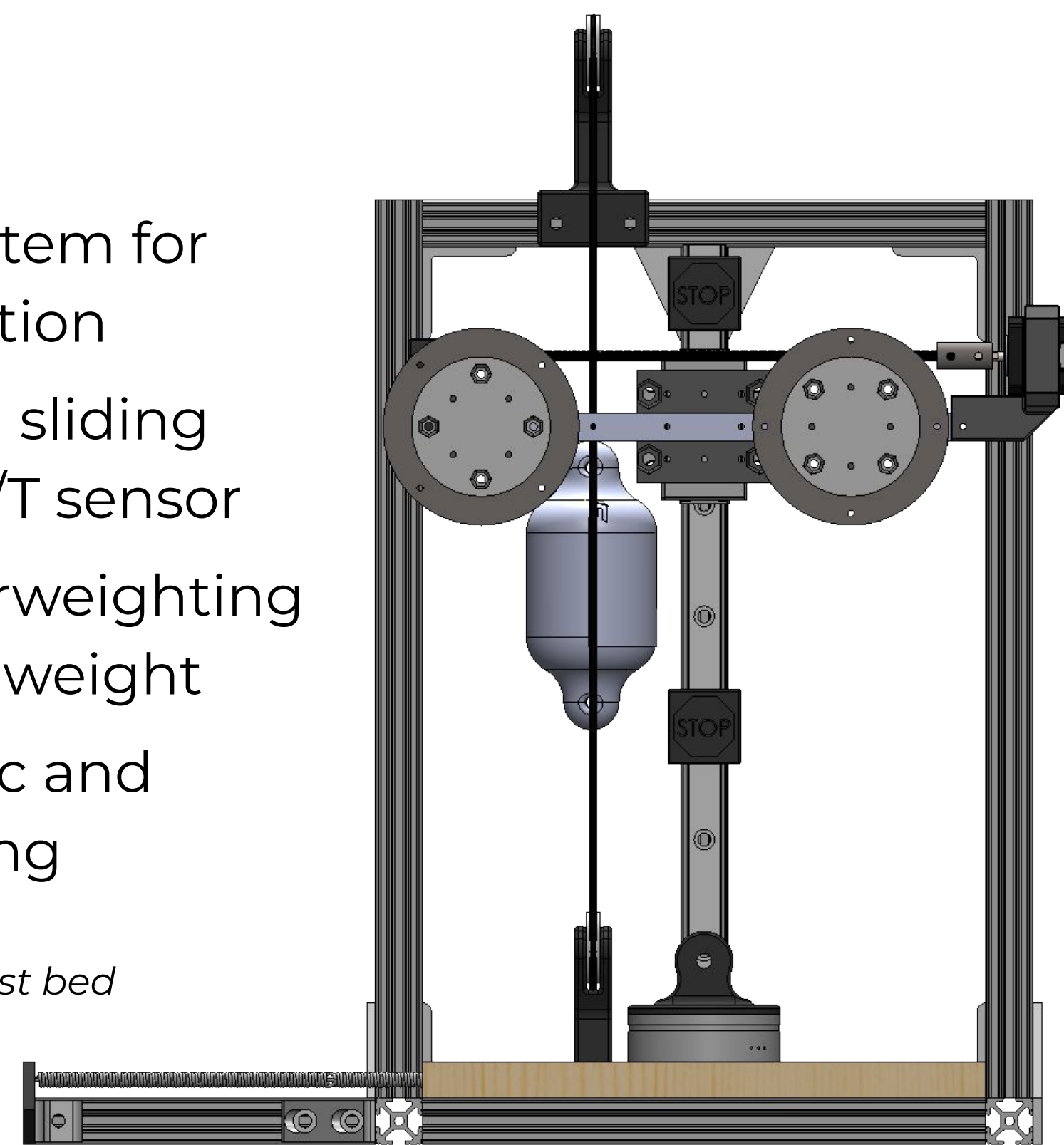
Fig 1: AI-generated rendering of robot (via Perplexity.AI)

Challenge: Create a linkage system that changes mechanical advantage through geometric transformation and allowing complex motion.
Solution: 5-bar pantograph leg with adjustable link lengths and ground link spacing, tested using a custom testbed with integrated force sensing.

Test Bed

- X-Y Gantry system for planar leg motion
- Spring-loaded sliding platform for F/T sensor
- Pulley counterweighting offsets gantry weight
- Supports static and dynamic testing

Fig 2: CAD model of test bed



5 Bar Pantograph Linkage

- Two rotational DOFs for gait control
- Link augmentation 3rd DOF for mode switching
- Enables transformation without sacrificing control
- Full range of rotational motion at joints
- Modular design for rapid configuration changes

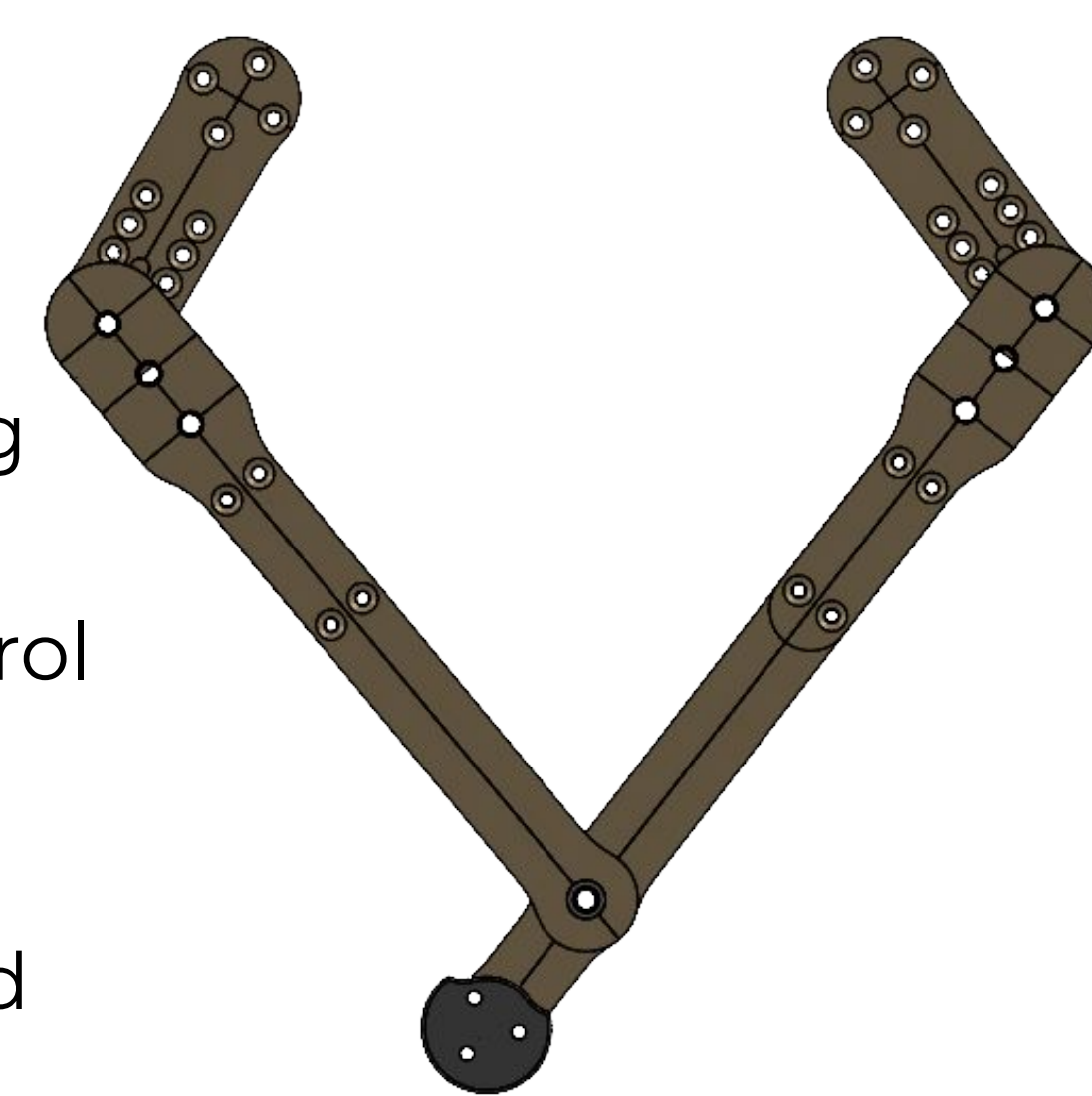


Fig 3: CAD model of 5 bar Pantograph Leg

Linear Actuator

- Stepper-driven lead screw to adjust length of ground link

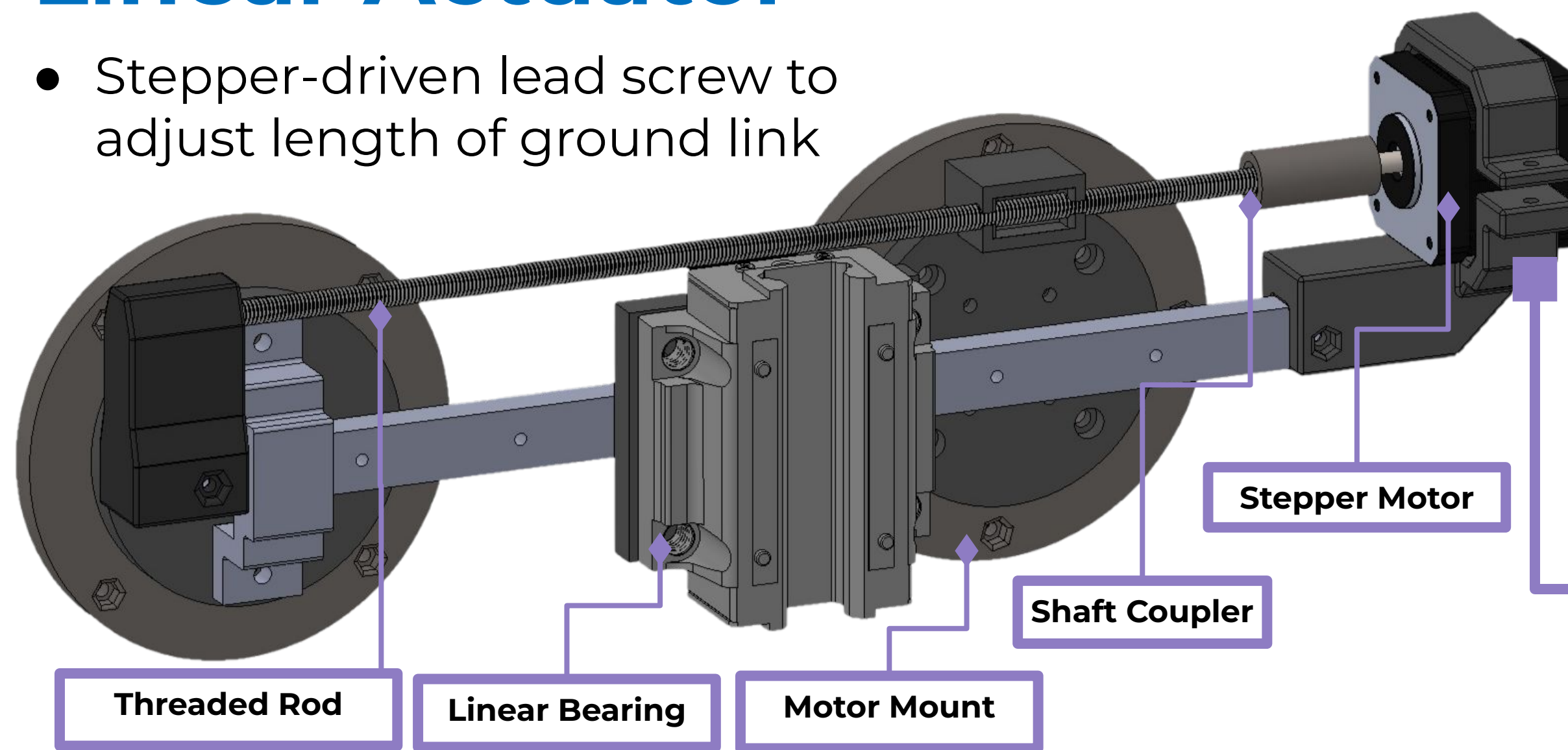


Fig 4: CAD model of linear actuator

Planetary Gearbox

- 5:1 Gear Ratio
- Increases torque while retaining necessary speed

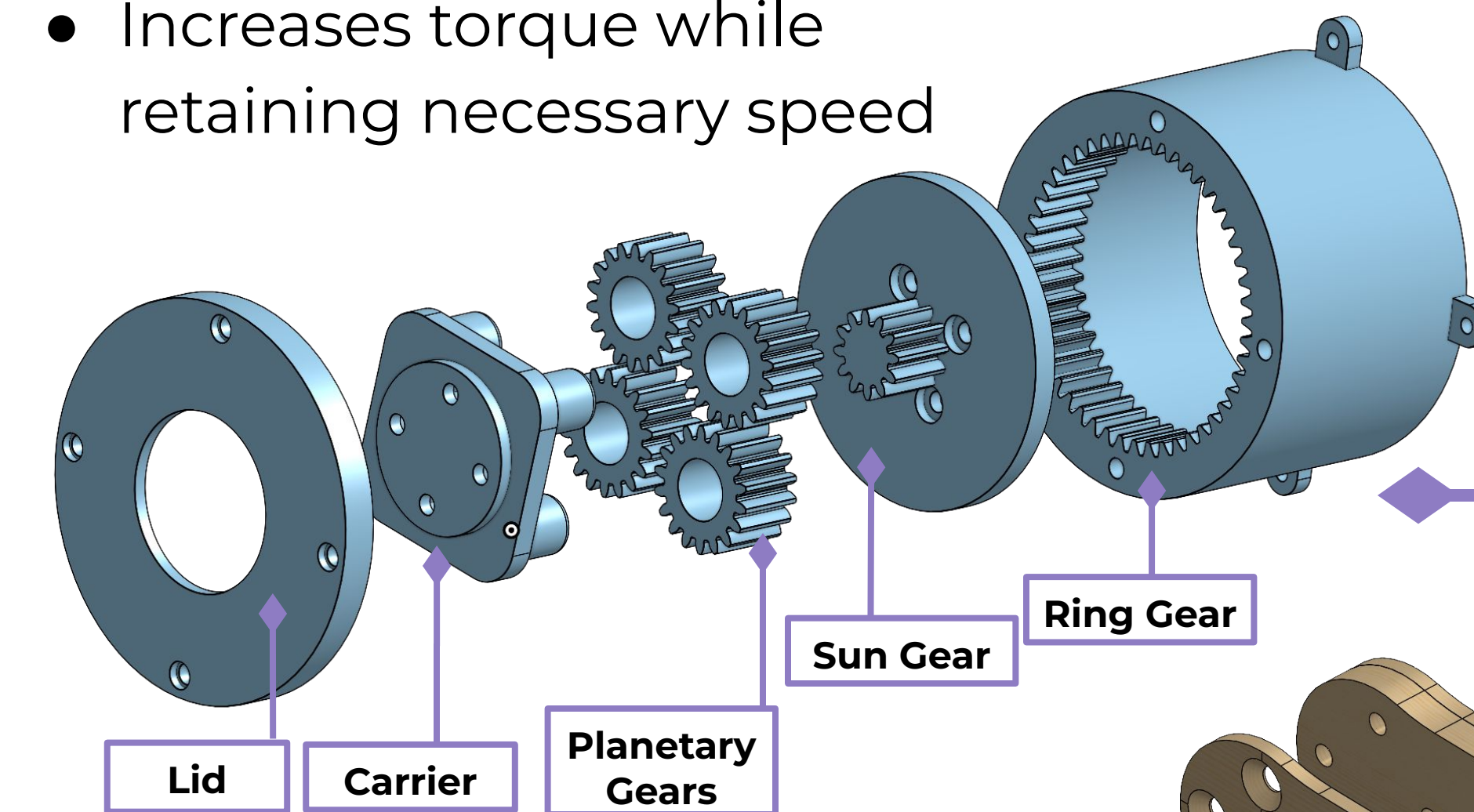


Fig 5: Exploded View of Gearbox CAD

Leg Links

- Sliding bearings (A/D) with multiple joint positions
- Adjustable length for configuration testing

Fig 6: CAD model of Link Assembly

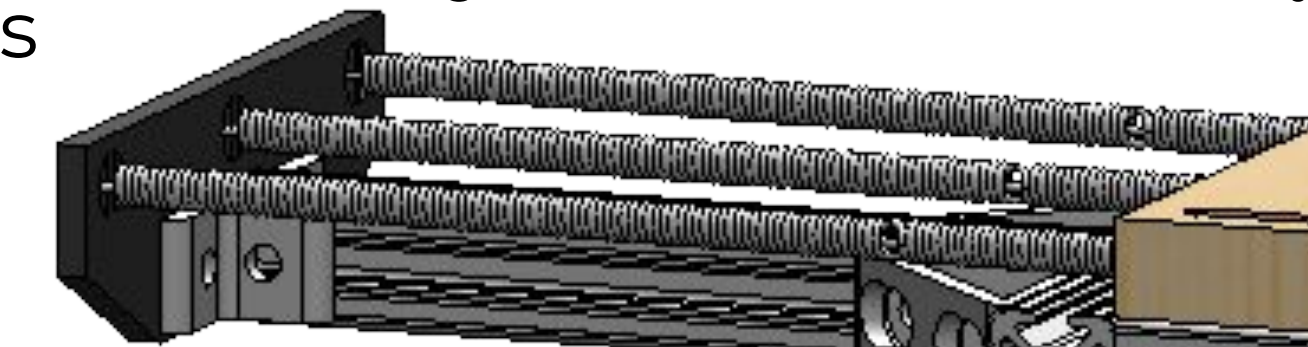
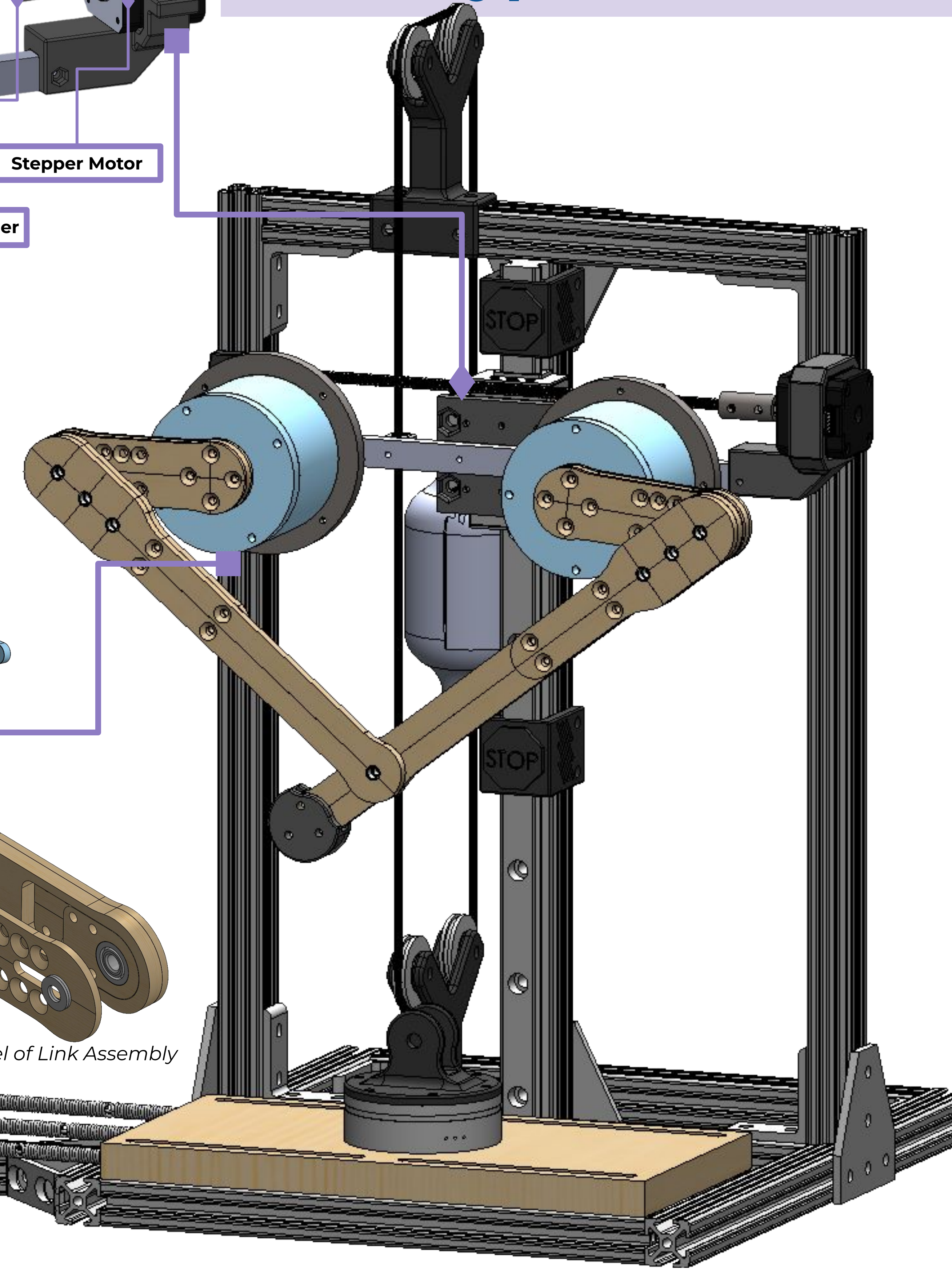


Fig 7: CAD model of Full System Assembly

5-Bar Pantograph Leg Prototype & Test Bed



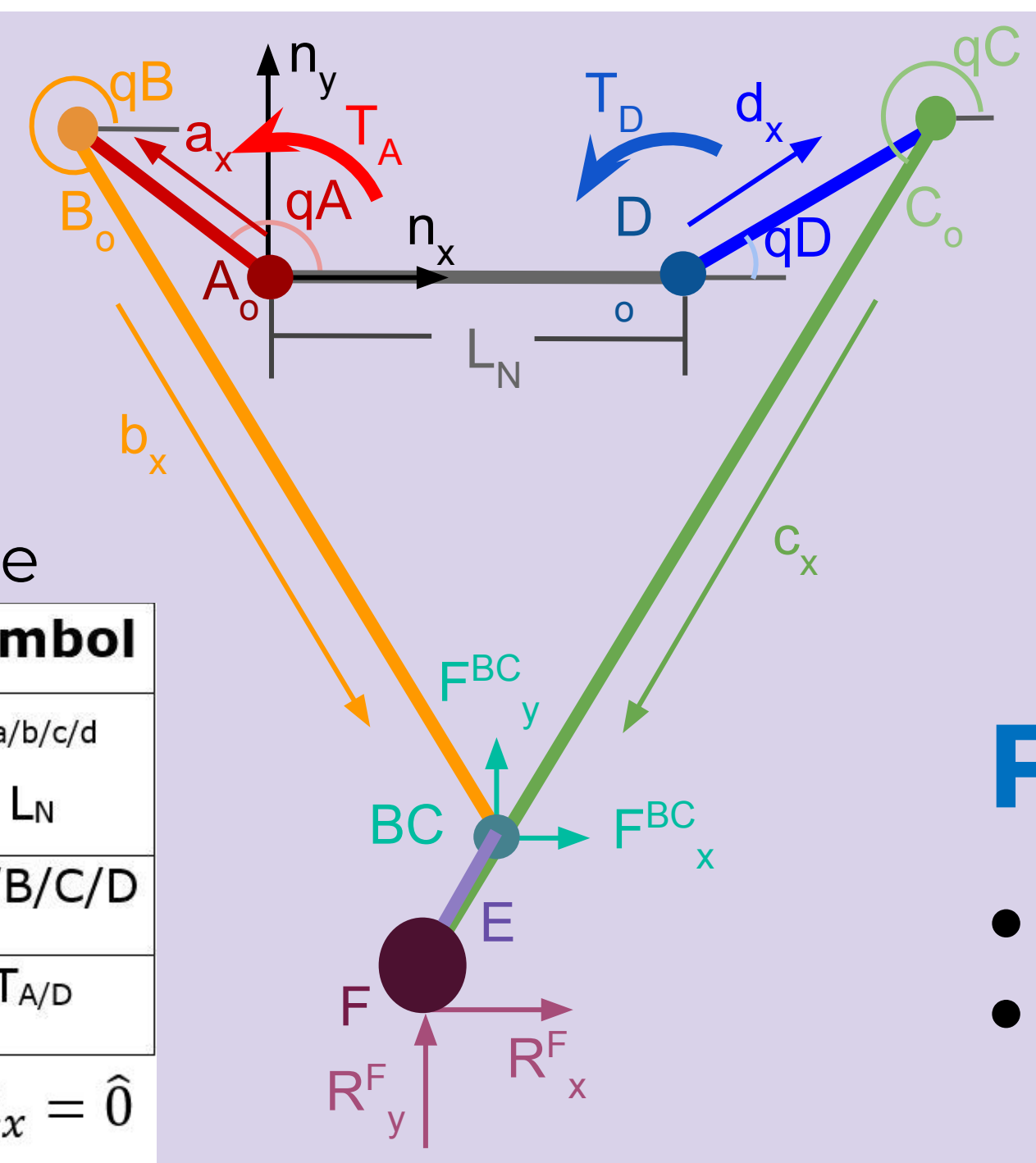
Linkage Setup

Assumptions:

- No friction
- Rigid uniform bars
- All links in the same plane

Quantity	Symbol
Length of link A/B/C/D	$L_{a/b/c/d}$
Length of ground link	L_N
Angle from $\hat{n}_x$ to a link in $+\hat{n}_z$	$q_{A/B/C/D}$
Torque applied on link A/D in $+\hat{n}_z$	$T_{A/D}$
$L_A \hat{a}_x + L_B \hat{b}_x - L_C \hat{c}_x - L_D \hat{d}_x - L_N \hat{n}_x = \hat{0}$	

Figures 8 and 9: Loop Equation (Left) and Diagram of Linkage Setup (Right)



Powered Components



Future Improvements

- Study asymmetrical length augmentation
- Optimize link lengths and gait profiles for various conditions of terrain
- Implement powered length adjustment for all links

Simulated Results

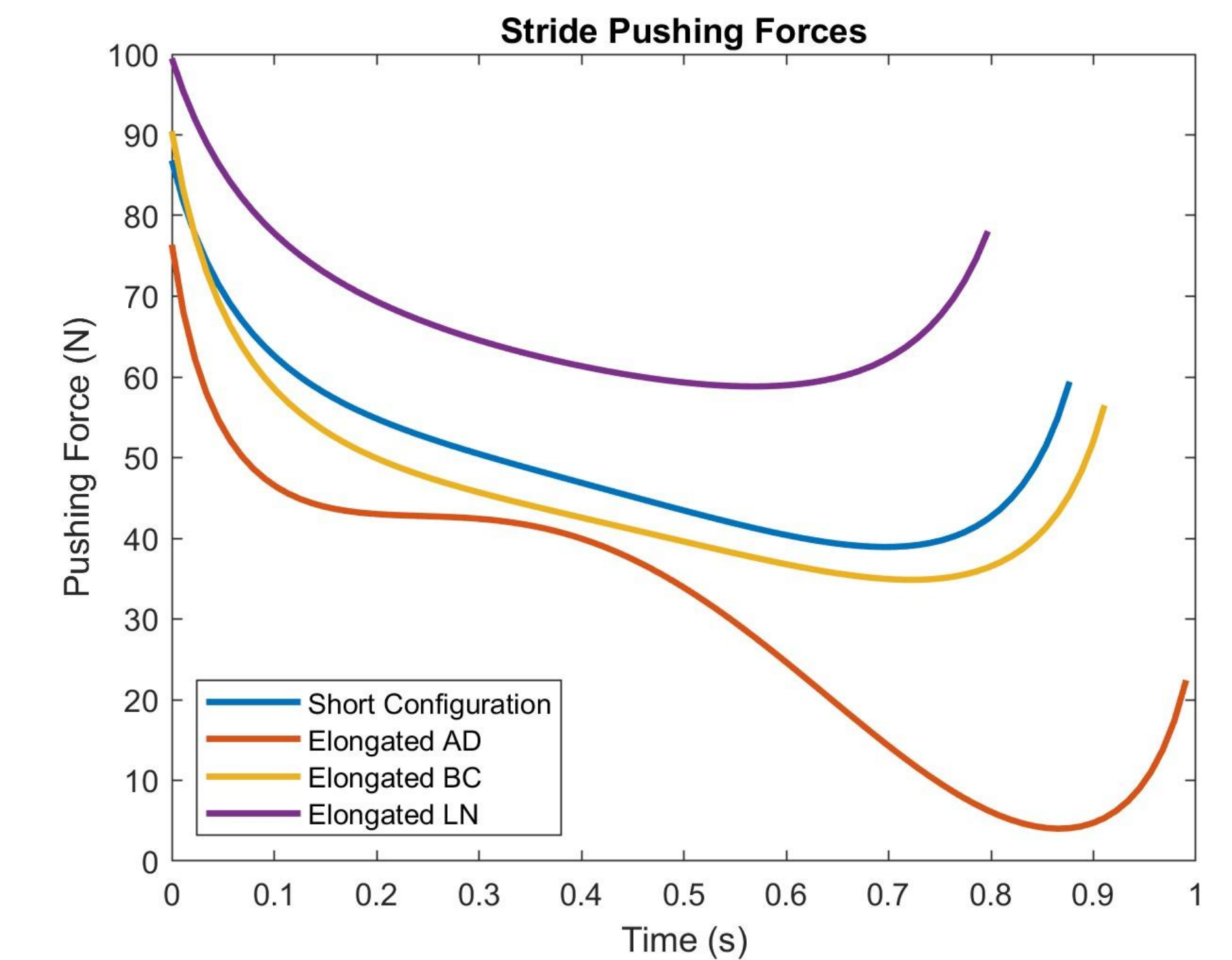


Fig 10: Comparison of stride pushing (horizontal) forces

- Dynamic analysis of forces through the stride of a walk cycle done using jacobian of the system
- Configurations that create larger strides reduce force output

Walk Cycle Comparison

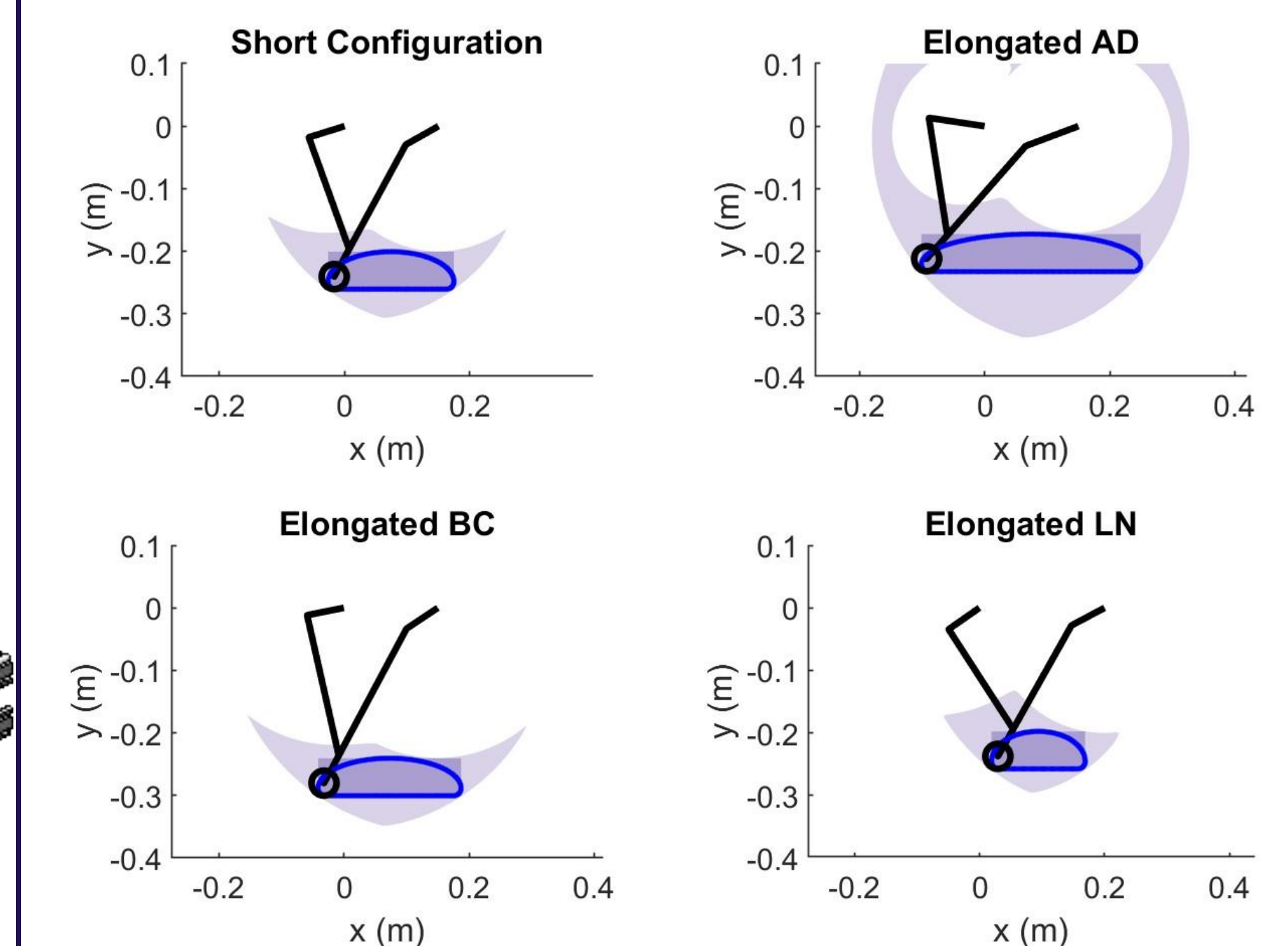


Fig 11: Walk cycle (blue) and reachable workspace (purple) for all four different configurations.

- Elongation of A, B, C, & D links increases max stride length while elongation of the ground link decreases it.
- Largest stride length of 35cm was achieved when the A & D links were elongated.

Impact on Society / Safety

- Enables casualty rescue in unsafe or inaccessible environments and reduces risks and physical strain for emergency personnel.
- In compliance with ASME, NFPA, ANSI, and ISO standards for safe mechanical and electrical design.

Acknowledgements

We thank Dr. Michael Yip, Elizabeth Peiros, and Calvin Joyce for their guidance and support as project sponsors.
We also appreciate the instruction and feedback from Dr. Huihui Qi and Dr. Nicholas Gravish throughout the course.

References

(1) B. Fallahi et al. A study of the workspace of five-bar closed loop manipulator, Mechanism and Machine Theory, 1994. (2) M.T. Hoang et al. Study and Development of Parallel Robots Based on 5-Bar Linkage, National Conference on Machines and Mechanisms, 2015. (3) J. Ma et al. Design, analysis and experiments of hexapod robot with six-link legs for high dynamic locomotion, Micromachines, 2022. (4) ATI Industrial Automation. ATI axia80 Force Torque Sensor Package, www.universal-robots.com/marketplace/products/01tP40000071NGXIA2/ (5) MyActuator. RMD-L-7015 product page, www.myactuator.com/l-7015-details (6) B&K Precision. 9115B-AT DC Power Supply, www.bkprecision.com/products/power-supplies/9115B-AT